

Final Project Tips

1. Project Selection and Scope

- You will independently choose your final project topic related to Natural Language Processing with a focus on **Urdu language (You can choose other topics also)**.
- Select a project clearly related to human language, such as text processing, understanding, generation, translation, sentiment analysis, or speech recognition specifically applied to Urdu.
- Your project should be **technically feasible** within the available timeframe and resources. Consider the availability of datasets, computational resources, and software tools.
- **The Department will soon provide multiple GPU resources**, which you can leverage to develop more computationally intensive models. Plan your project to optimally use these resources.
- Aim for projects that deliver concrete outcomes such as working models, new datasets, evaluation metrics, or analysis reports.
- Projects should be non-trivial and ideally push beyond replication by providing improvements, new applications, or insights in Urdu NLP.

2. Defining a Clear Question or Goal

- Start with a **specific guiding question or hypothesis** about Urdu language NLP. Examples:
 - “Does fine-tuning pre-trained Urdu BERT improve sentiment classification accuracy compared to training from scratch?”
 - “Can a rule-based approach improve Urdu part-of-speech tagging on noisy text?”
 - “How does dialectal variation affect Urdu speech-to-text systems?”
- Avoid vague or overly broad questions without measurable outcomes.

3. Evaluation and Metrics

- Choose **clear, objective metrics** for assessing your project’s success (e.g., accuracy, F1 score, BLEU for translation, or human evaluation with a rubric).
- Prepare evaluation rubrics if human assessment is used to ensure consistency.

- Aim for results that enable comparison of different methods or analysis of trends, not just system building without evaluation.

4. Creative and Scientific Contribution

- Build on existing methods but push beyond simple re-implementation. Consider improvements, ablations, or new applications in Urdu NLP.
- Negative or inconclusive results are acceptable if you analyze and explain them well.
- Projects may also develop new datasets, benchmark models, or explore theoretical insights specific to Urdu language processing.

5. Resource and Time Management

- Scope projects to fit within the available semester/time frame.
- Be realistic with computing resources; now you can utilize **the upcoming GPU resources recently made available by the Department**, which can enable training of more sophisticated models.
- Use open-source tools and pre-trained models when appropriate.
- It's better to complete a smaller, well-executed project than an overly ambitious one.

6. Documentation and Reporting

- Submit a **clear project proposal** early, describing your question, methodology, data sources, and evaluation plan.
- Your final report should clearly present your motivation, experimental setup, results, analysis, and reflections on lessons learned and future work.

7. Recommended Topics (Urdu Language Focus)

- Urdu text classification (sentiment, topic, spam detection)
- Named entity recognition for Urdu
- Machine translation between Urdu and other languages
- Urdu speech recognition or synthesis
- Urdu question answering or information retrieval
- Dialect or script variation in Urdu NLP

- Building or improving Urdu language datasets or lexicons

8. Additional Resources and Inspiration

- For more inspiration and examples of successful projects, visit the **Stanford CS224n course project page**: <https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1214/project.html>.
- This page includes a variety of project summaries from previous years covering diverse NLP topics that can spark ideas tailored to Urdu language projects.

This guidance is designed to help you successfully choose and complete meaningful, feasible, and impactful projects in NLP course, making use of the new GPU resources and drawing inspiration from leading NLP coursework and projects.

If you have questions or need feedback on your project ideas, please reach out early.

Good luck with your projects!

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